

CHAPTER 1



Workshop on Essentials of Mortality Investigation of Asian Elephant

Illustrated Notes · GIS Maps · Data Analysis

5th–6th June 2026

Dharamjaigarh & Raigarh Van Mandals · Bilaspur Forest Circle · Chhattisgarh

Organised by: Chhattisgarh Forest Department

Technical Support: Wildlife Institute of India, Dehradun

Laboratory: ICAR-IVRI, Izatnagar · NDVSU, Jabalpur

1. Workshop Overview & Objectives

1.1 Background

This two-day workshop was convened by the **Chhattisgarh Forest Department** with technical support from **Wildlife Institute of India (WII), Dehradun** to build field-level capacity in elephant mortality investigation across the Bilaspur Forest Circle. **84 participants** attended from the Forest Department and Veterinary Service Department of Chhattisgarh State.

1.2 Objectives

- **Biology, Behaviour & Disease** — Understand elephant biology, behaviour, and the major infectious and non-infectious diseases causing mortality.
- **Necropsy, Biosafety & Sampling** — Hands-on competency in field necropsy, tissue sampling, cold-chain documentation, and PPE protocols.
- **Surveillance, Disposal & Forensics** — Operational frameworks for carcass disposal, site decontamination, standardised forensic reporting and health monitoring.

1.3 Resource Persons

Name	Designation	Institution	Expertise
Dr. A. B. Shrivastav	Founder Dir., School of Wildlife Forensic & Health	NDVSU, Jabalpur	Wildlife Health, Forensics, Pathology
Dr. Parag Nigam	Sr. Scientist & Head, Wildlife Health	WII, Dehradun	Chemical Capture, Conservation Medicine
Dr. Karikalan Mathesh	Sr. Scientist, CWCMDs	ICAR-IVRI, Bareilly	Wildlife Pathology, Histopathology, Molecular Dx
Dr. Tapendra Saini	Scientist C	WII, Dehradun	Animal Genetics, Conservation Breeding
Dr. C. P. Sharma	PTO & In-charge, Morphology Unit	WII, Dehradun	Wildlife Forensics, Crime-Scene Mgmt

2. Elephant Population — Chhattisgarh (2001–2026)

The elephant population has grown from **24 individuals (2001)** to an estimated **451 elephants (2026)** — a CAGR of ~13% over 25 years. The 2021–2026 study period alone saw a **+62% increase (279 → 451)** at ~10% CAGR, driven by natural dispersal from Odisha and Jharkhand into Bilaspur Circle.

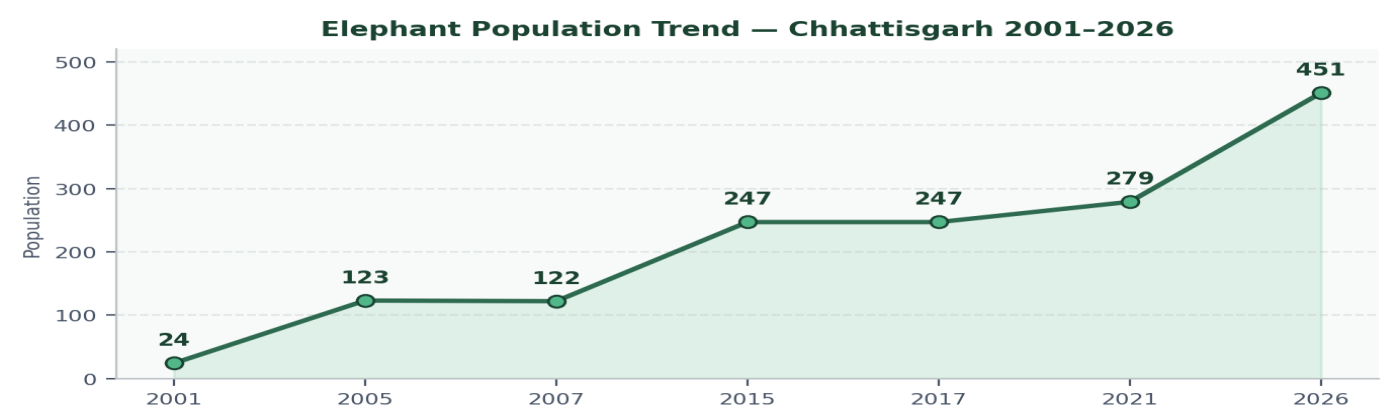


Figure 1: Population trend 2001–2026. Source: WII Census & CG Forest Dept. Estimates.

Year	Count	Notes
2001	24	Founding dispersal from Odisha/Jharkhand
2005	123	First major census count (+412% in 4 yrs)
2015	247	Significant consolidation
2021	279	Official Census — 10% CAGR since 2005
2026	~451 est.	Net +172 over 5-year study period

Bilaspur Circle hosts the highest elephant density in CG. 2025-26 recorded the highest single-year human-elephant conflict events in the 5-year study period.

3. Elephant Casualties — Year-wise Overview (2021–2026)

49 confirmed elephant deaths over the study period. Dharamjaigarh (DH) contributes 27 deaths and Raigarh (RG) 22. The year 2025-26 is the **record year (12 deaths)**, driven by a sharp RG spike. The 2026-27 cohort (5 deaths in 3 months — all calves, all drowning) signals an accelerating acute threat.

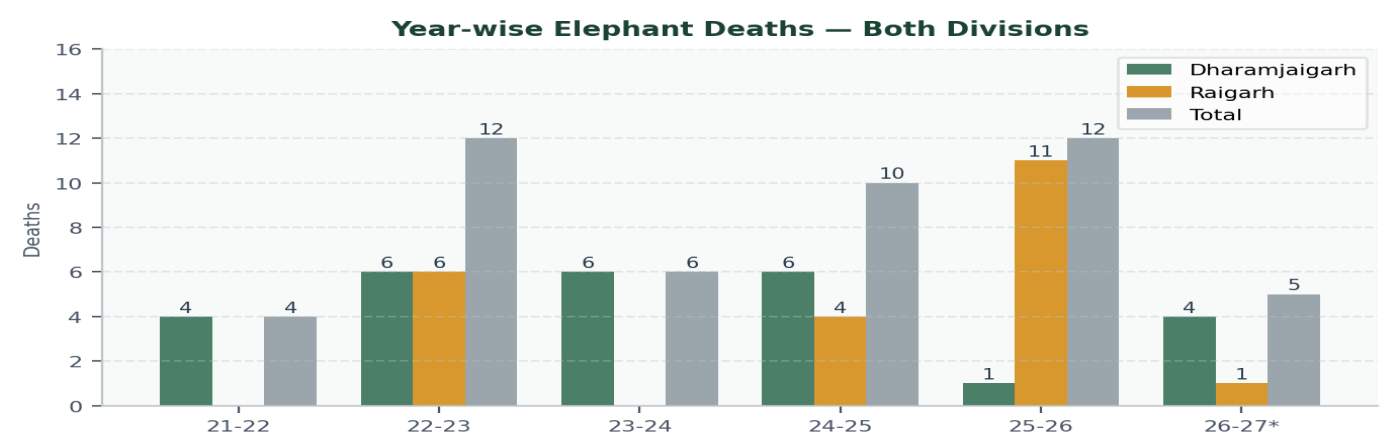


Figure 2: Year-wise deaths by division — Dharamjaigarh (green), Raigarh (amber), Total (grey).

Year	DH	RG	Total	Notable Pattern
2021-22	4	0	4	All DH — founding year of RG dispersal
2022-23	6	6	12	RG emerges; multi-cause cluster
2023-24	6	0	6	DH dominant; RG quiet
2024-25	6	4	10	Both divisions active; mixed causes
2025-26	1	11	12	★ RECORD — RG electrocution & drowning surge
2026-27*	4	1	5	All calves, all drowning — ongoing
TOTAL	27	22	49	5+ year study

■ **2026-27 early signal: all 5 deaths in first 3 months are calves (0–2 yr) from suspected drowning. Immediate dry-season intervention required.**

4. Cause of Death Analysis

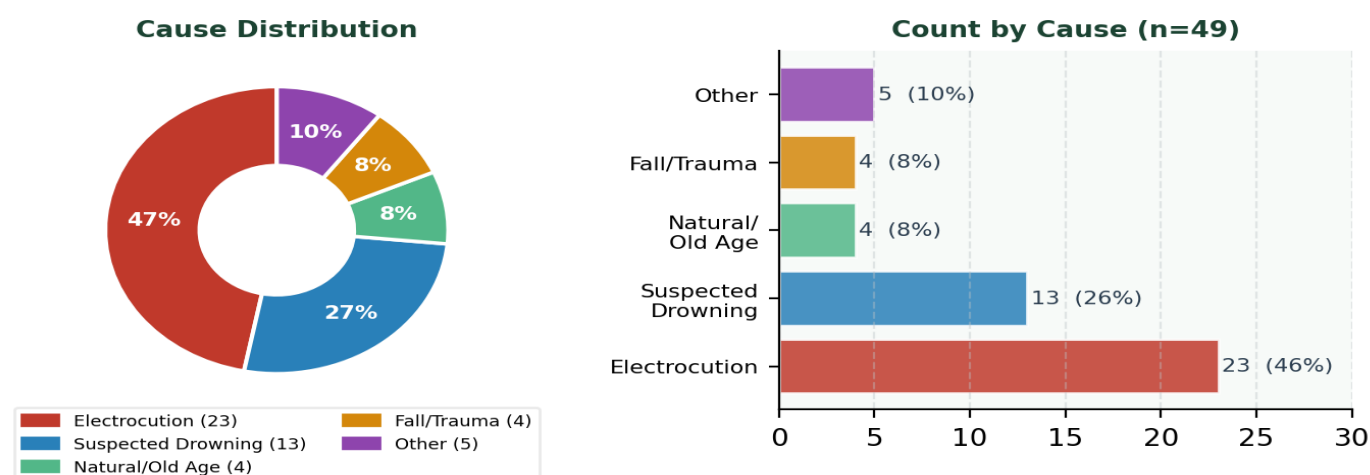


Figure 3: Cause distribution — donut (left) and count with percentages (right). n = 49.

Cause	Count	%	Key Risk Factor
Electrocution	23	47%	Illegal/low-hung power lines near crop fields; 3 multi-simultaneous events
Suspected Drowning	13	27%	Dams, ponds, seasonal nalas; 92% calves/young
Natural / Old Age	4	8%	Senescence; female-skewed (4F:1M)
Fall / Trauma	4	8%	Hills, nalas, infighting
Other (heat/sun/unknown)	5	10%	Heatstroke, cardiovascular shock, weakness

Electrocution (47%) + Drowning (27%) = 74% of deaths — both fully preventable with targeted infrastructure, enforcement, and field monitoring.

5. Demographic Analysis — Age & Seasonal Profiles

5.1 Age Group Distribution

50% of all deaths are calves (0–2 yr). With juveniles included, **61% of deaths occur below 15 years of age** — a direct long-term threat to population viability and recruitment.

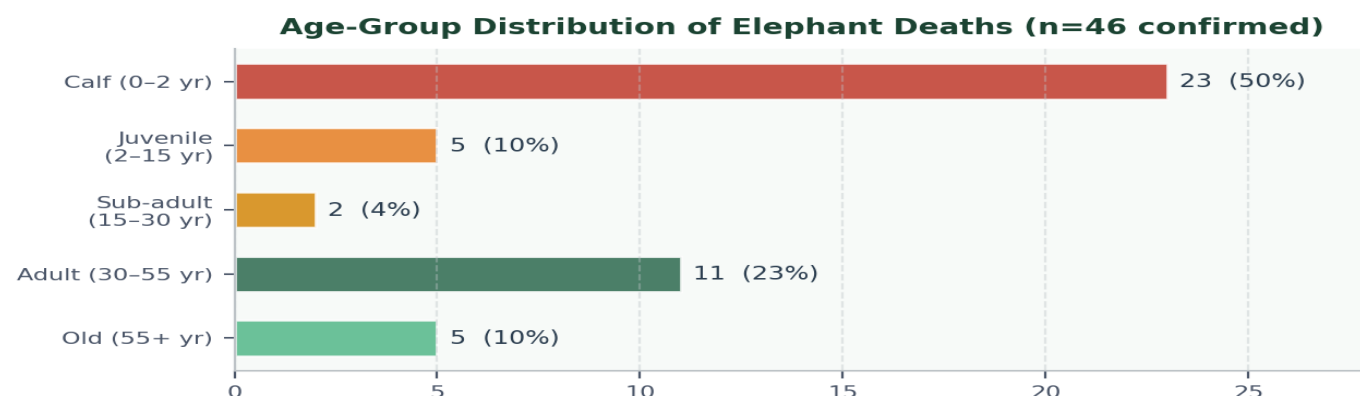


Figure 4: Age-group distribution. Calf class alone = 50% of all deaths.

5.2 Seasonal Variation

October is the deadliest month (8 deaths), driven by electrocution during crop-harvest electric fencing. Oct–Dec = 42% of all annual deaths. A secondary drowning peak occurs Jan–May when dry-season water levels are low and banks are steep.

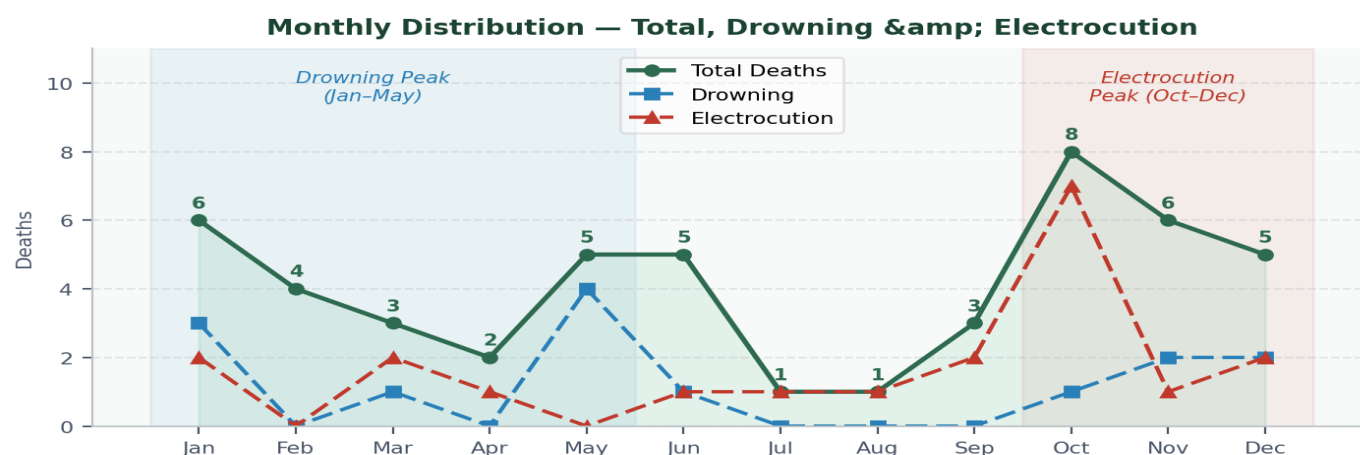


Figure 5: Monthly deaths — total (green), drowning (blue), electrocution (red). Shaded zones mark seasonal peaks.

Season	Drowning	Electrocution	Driver
Jan–May (Dry)	8 (62%)	6 (30%)	Low water, steep banks; crop fields quiet
Jun–Sep (Monsoon)	1 (8%)	5 (25%)	Nalas swell; some fencing active
Oct–Dec (Harvest)	4 (31%)	10 (50%)	Crop protection fencing at maximum

6. Range-wise Distribution

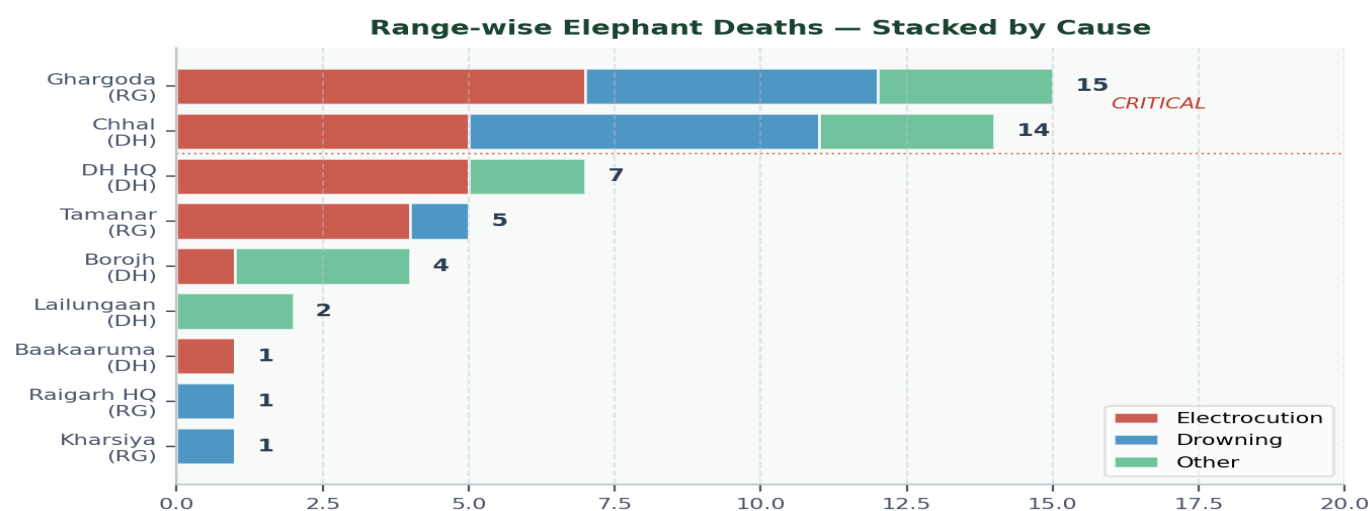


Figure 6: Range-wise stacked deaths — electrocution (red), drowning (blue), other (green). Chhal + Ghargoda = CRITICAL zones.

Range / Beat	Division	Total	Risk	Elec	Drown	Other
Ghargoda	Raigarh	15	CRITICAL	7	5	3
Chhal	Dharamjaigarh	14	CRITICAL	5	6	3
Dharamjaigarh HQ	Dharamjaigarh	7	HIGH	5	0	2
Tamanar	Raigarh	5	MODERATE	4	1	0
Borojh	Dharamjaigarh	4	HIGH	1	0	3
Lailunga	Dharamjaigarh	2	HIGH	0	0	2
Baakaaruma	Dharamjaigarh	1	MODERATE	1	0	0
Raigarh HQ	Raigarh	1	LOW	0	1	0
Kharsiya	Raigarh	1	LOW	0	1	0

PRIORITY: Chhal (DH) + Ghargoda (RG) = 29 deaths (59% of all elephant casualties). These two ranges require the most urgent multi-agency intervention.

7. GIS Analysis — Casualty Geo-mapping

7.1 Master Casualty Map

All 49 elephant deaths and 45 human deaths geo-referenced and plotted. Elephant deaths are colour-coded by cause (electrocution = red, drowning = blue, natural = green, trauma = amber). Human deaths are shown as triangles. The division boundary (dashed) separates Dharamjaigarh (west) from Raigarh (east).

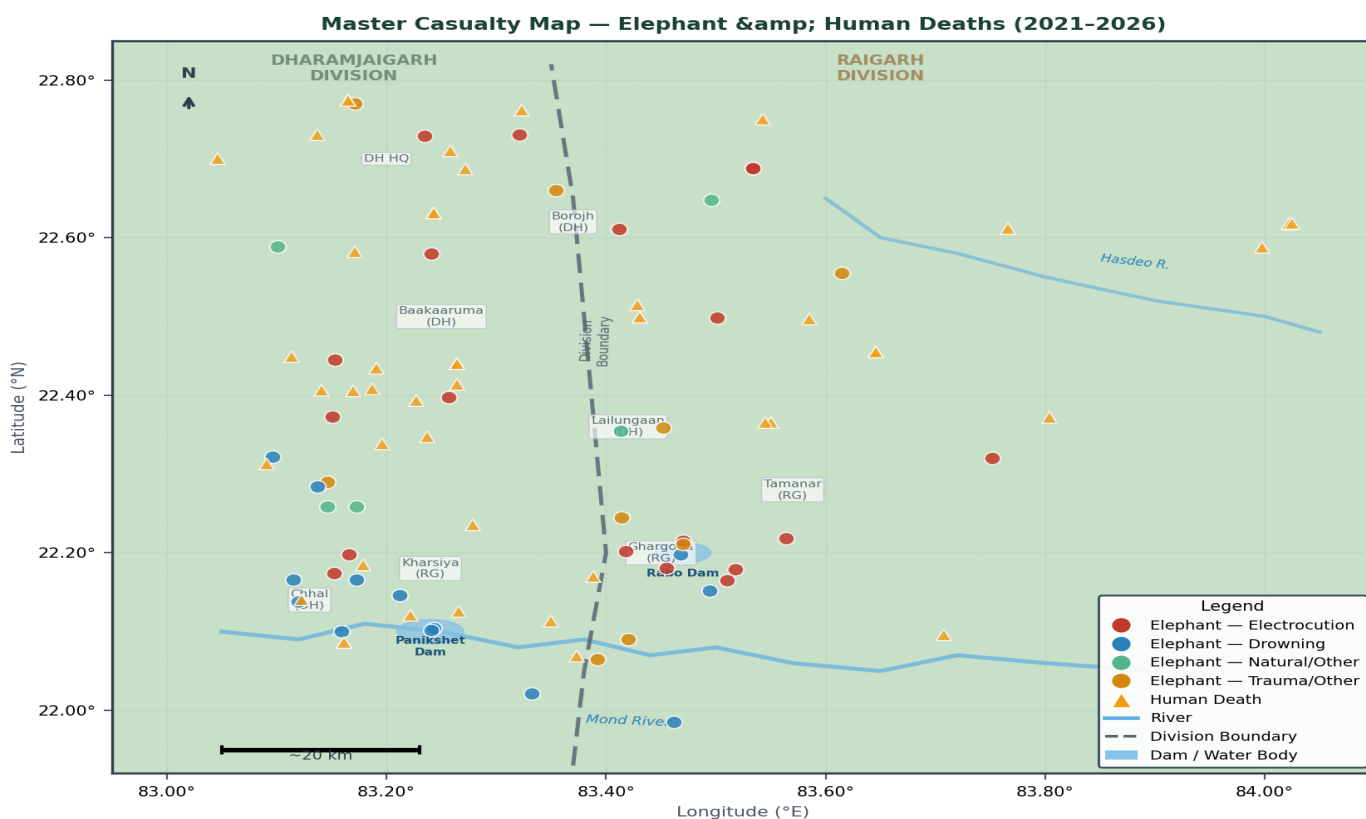


Figure 7: Master casualty map — elephant deaths (circles, cause-coded) + human deaths (triangles, orange). All GPS-referenced mortality events 2021–2026.

Key spatial observations:

- The western cluster (DH, lon 83.05–83.40) is dominated by **electrocution deaths** in the Chhal, Borogh and Dharamjaigarh HQ ranges.
- The central-eastern cluster (RG, lon 83.40–83.60) shows a **mixed drowning-electrocution pattern** concentrated around Ghargoda range and the Panikshet/Rabo dam corridor.
- Human deaths mirror elephant corridors, indicating shared forest-agriculture interface zones.

8. GIS — Elephant Deaths: Cause & Spatial Pattern

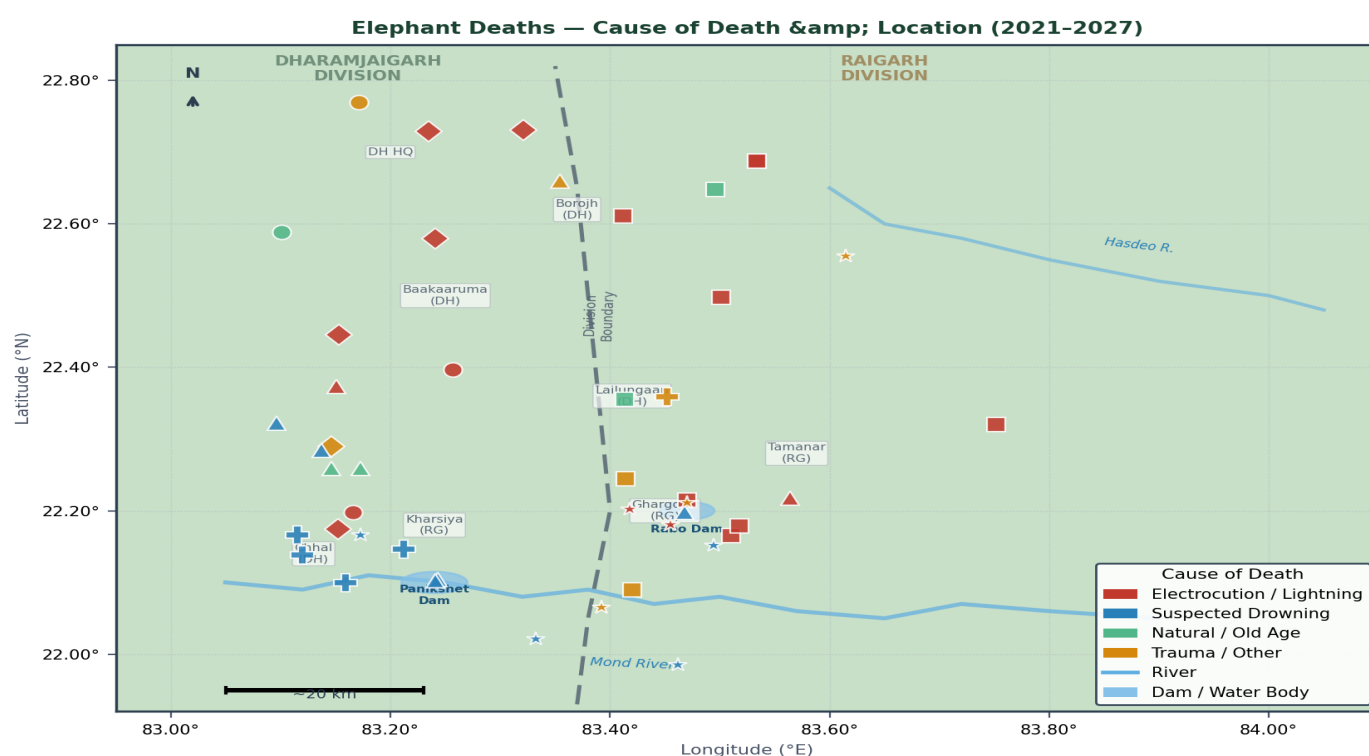


Figure 8: All 49 elephant deaths — colour coded by confirmed cause of death. Red = electrocution, Blue = drowning, Green = natural, Amber = trauma/other.

8.1 Spatial Interpretation

- **Electrocution cluster (red):** Concentrated in DH west (Chhal, DH HQ) and RG central (Ghargoda, Tamanar) — correlates with high-density agricultural zones and illegal crop-fence corridors.
- **Drowning cluster (blue):** Mainly RG division near Panikshet and Rabo dams, and Mond River nalas (DH south). All dry-season water trap locations.
- **Natural/old age deaths (green):** Scattered broadly — no geographic hotspot pattern.
- **Trauma/other (amber):** Sparse and scattered — no dominant cluster.

9. GIS — Year-wise Progression of Elephant Deaths (2021–2027)

Year-wise Elephant Death Progression (2021–2027)

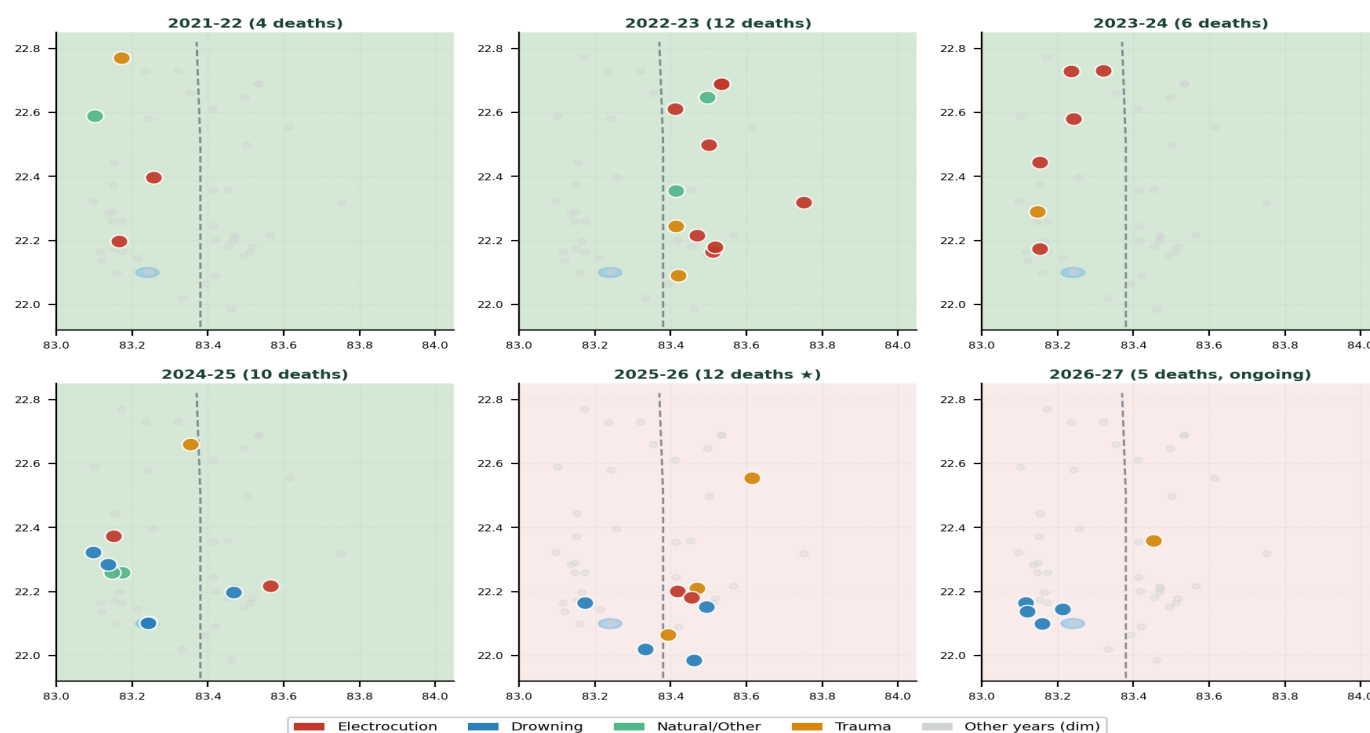


Figure 9: Year-wise small-multiple maps. Active year deaths shown in full colour; all other years dimmed. Red background = record/critical years (2025-26, 2026-27).

9.1 Key Spatial Trends by Year

- **2021-22:** Deaths confined to DH division only — all western half. RG corridor not yet active.
- **2022-23:** First RG deaths appear — electrocution cluster in central RG. Largest single-year count (12).
- **2023-24:** Returns to DH dominance — electrocution hotspot intensifies in DH south.
- **2024-25:** Both divisions active. First confirmed drowning cluster in RG dam zone.
- **2025-26 (RECORD):** Dramatic RG surge — 11 of 12 deaths are in RG, concentrated around Ghargoda and dam corridor. DH drops to 1 death.
- **2026-27 (ongoing):** 4 of 5 deaths in DH south — all calves, all drowning. New cluster emerging around Mond River nalas.

10. GIS — Drowning Hotspot Analysis

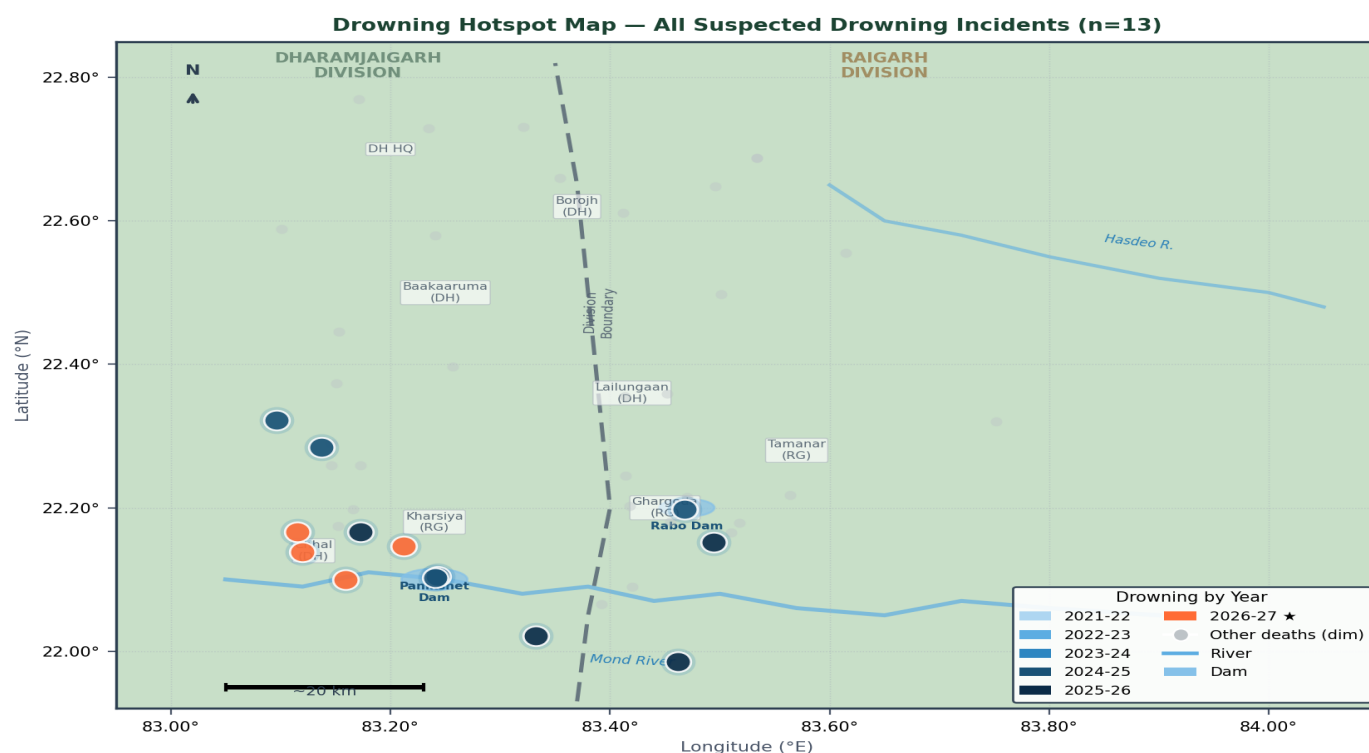


Figure 10: Suspected drowning incidents (n=13) year-coded from light blue (2021-22) to orange (2026-27). Non-drowning deaths are dimmed. Note clustering around dam zones and river nalas.

10.1 Drowning Incident Locations

Location / Water Body	Division	Year	Coordinates	Notes
Panikshet Dam	Raigarh	2024-25	22.10°N 83.24°E	2 calves, consecutive
Rabo Dam	Raigarh	2024-25	22.20°N 83.47°E	1 calf
Panikshet Dam (repeat)	Raigarh	2024-25	22.10°N 83.24°E	1 calf
Mond River nala (DH S)	Dharamjaigarh	2025-26	22.17°N 83.17°E	Arsenic area
Ghargoda nala (RG)	Raigarh	2025-26	Multiple	4 calves cluster
Mond River (Kharsia)	Raigarh	2026-27	22.14°N 83.12°E	Pelvic fracture noted
Gurda River (DH S)	Dharamjaigarh	2026-27	22.04°N 83.13°E	Gurda case study
Pond (DH south)	Dharamjaigarh	2026-27	22.17°N 83.12°E	2026-27 cohort

ARSENIC DETECTION: Tamnar Range (Raigarh Division), December 2025 — arsenic found in visceral samples from 2 specimens near Rabo/Panikshet dam corridor. Environmental water sampling of these catchments is urgently required.

11. GIS — Electrocution Hotspot Analysis

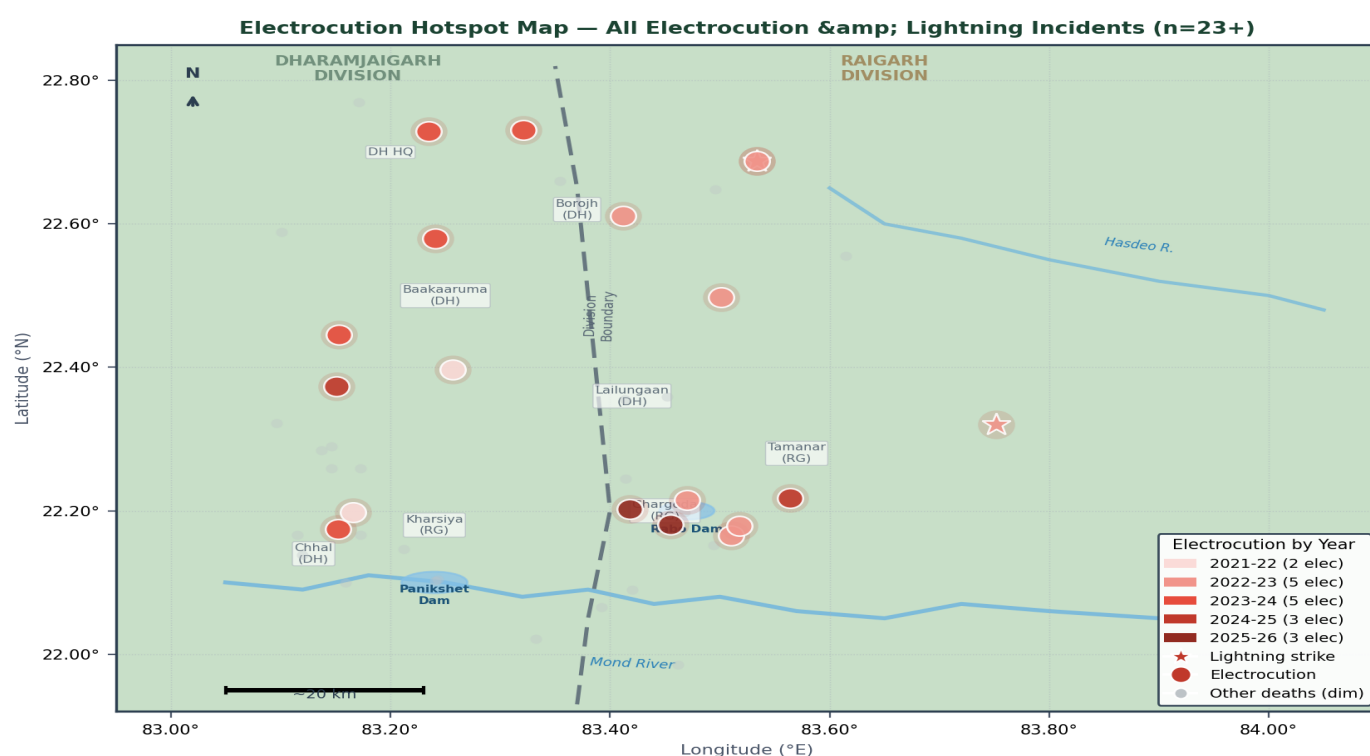


Figure 11: All electrocution + lightning deaths — year-coded from pale red (2021-22) to dark red (2025-26). Stars = lightning strikes; circles = electrocution. Non-electrocution deaths dimmed.

11.1 Electrocution Spatial Patterns

- **DH western cluster:** Chhal range (DH south) — 5 electrocution deaths across multiple years, indicating persistent illegal low-hung lines in the Koylar-Chhal agricultural corridor.
- **DH HQ cluster:** 5 electrocution deaths in Dharamjaigarh HQ area — crop-field perimeter fencing in active cultivation zones.
- **RG central cluster:** Ghargoda and Tamanar ranges — 7+4 = 11 electrocution deaths, heaviest concentration; Ghargoda is the single deadliest range.
- **Lightning strikes (2022-23):** Two in DH east (lon 83.45–83.75) — isolated events.
- **Peak season Oct–Dec** (harvest) is when most electrocution events cluster.

12. GIS — Human Casualty Map (HEI Deaths 2021–2026)

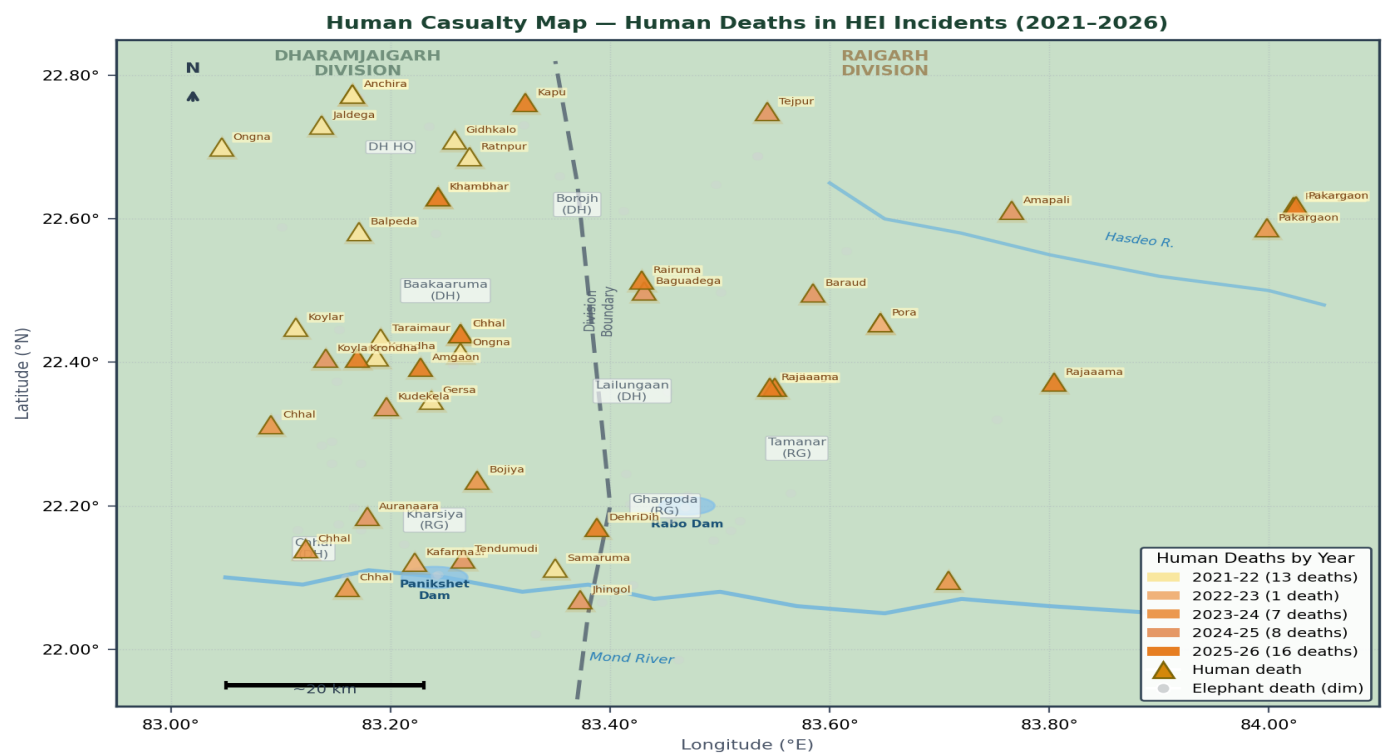


Figure 12: Human deaths from human-elephant interface (HEI) incidents — year-coded triangles. Elephant deaths shown dimmed for spatial reference. Total 45 human deaths across both divisions.

12.1 Human Casualty Spatial Notes

- **DH division (36 human deaths):** Concentrated in the Chhal-Borojha-Lailunga corridor (DH south-west) — same zone with high elephant electrocution deaths, confirming shared forest-agriculture interface.
- **RG division (9 human deaths):** Scattered across Ghargoda-Tamanar area; lower human casualties despite higher elephant mortality.
- **2021-22 peak (13 human deaths) and 2025-26 peak (16 human deaths)** correlate with years of high elephant movement and crop-harvest season overlap.

Division	Human Deaths	Elephant Deaths	Deadliest Range (Human)
Dharamjaigarh	36	27	Chhal (10) > Borojha (9) > Lailunga (8)
Raigarh	9	22	Ghargoda (5) > others

13. Post-mortem & Laboratory Analysis

13.1 PM Cause Distribution (41 confirmed PM records)

Cause	PM-Confirmed	%
Electrocution → Cardio-Respiratory Failure	~20	49%
Suspected Drowning	~9	22%
Natural / Old Age	~5	12%
Trauma / Fall	~5	12%
Other / Undetermined	~2	5%

13.2 Drowning — Consistent PM Findings (all 13 cases)

PM Finding	Frequency
Frothy nasal/oral discharge — emphysema aquosum	ALL CASES
Lungs waterlogged, heavy, congested, oedematous	ALL CASES
Water in trachea & bronchi	ALL CASES
Gastric content — large volume of water & plant debris	ALL CASES
Skin maceration / hide wrinkling	CONSISTENT
Petechial haemorrhages — conjunctiva / sub-pleural	VARIABLE
Diatom test POSITIVE — ante-mortem drowning confirmed	ALL TESTED CASES

13.3 Histopathology

- Pulmonary oedema — alveolar flooding with eosinophilic fluid.
- Pulmonary haemorrhage — interstitial and alveolar spaces.
- **Hepatocyte degeneration** in 3 cases — Tamnar Range, December 2025.
- Renal congestion — passive vascular engorgement.

13.4 Laboratory Results (ICAR-IVRI + NDVSU, 5 batches)

Test / Agent	Result	Note
Arsenic	DETECTED ■	2 specimens, Tamnar Range RG, Dec 2025 — elevated in viscera
EEHV-I PCR	NEGATIVE	Confirmed by ICAR-IVRI & NDVSU Jabalpur — viral aetiology ruled out
Nitrate-Nitrite	NEGATIVE	All 5 batches
Organophosphates / Organochlorines	NEGATIVE	All 5 batches
Heavy metals (Pb, Hg, Cd)	NEGATIVE	All 5 batches
Hydrogen Cyanide (HCN)	NEGATIVE	All 5 batches

ARSENIC: Environmental sampling of Rabo & Panikshet dam catchments recommended immediately. Results to be shared with CG Pollution Control Board for enforcement action.

14. Case Study — Gurda Elephant Calf Drowning · 01 June 2026

Date	01 June 2026
Location	Gurda, Kharsia Range, Raigarh Van Mandal
GPS	22.0439°N, 83.1313°E (22°3′16″N 83°8′11″E)
Elevation	256.04 ± 9.59 m
Water Body	Gurda River / exposed sandbank
Species	Asian Elephant (Elephas maximus)
Age Class	Calf (0–2 yr)
Cause	Suspected Drowning

Calf found on an exposed sandbank adjacent to the Gurda River. Photographic documentation at 10:56 hrs (aerial view) and 11:03 hrs (carcass, ground level). Incident is consistent with the established **dry-season drowning pattern** — low water levels and steep banks trap calves that approach the edge to drink.

Field note (original): — Gurda , ,

This case exemplifies the 2026-27 cohort: all 5 deaths are calves, all suspected drowning, all in first 3 months. Gurda is the southernmost recorded drowning event in Raigarh Division (22.04°N).

15. Key Findings, Recommendations & Action Points

15.1 Summary Statistics

Metric	Value
Study Period	2021-22 to 2026-27 (ongoing)
Total Elephant Deaths	49 confirmed
Total Human Deaths (HEI)	45 confirmed
Leading Cause (Elephant)	Electrocution — 23 deaths (47%)
2nd Cause (Elephant)	Suspected Drowning — 13 deaths (27%)
Most Vulnerable Age	Calves 0–2 yr — 23 deaths (50%)
Record Year	2025-26 — 12 deaths
2026-27 Signal	5 deaths in first 3 months — all calves, all drowning
CRITICAL Ranges	Chhal (14) + Ghargoda (15) = 59% of all elephant deaths
Drowning Season Peak	Jan–May = 62% of drowning deaths
Electrocution Season Peak	Oct–Dec = 50% of electrocution deaths
Arsenic Finding	Positive — 2 specimens, Tamnar Range, Dec 2025
EEHV Status	Negative — all batches tested
Toxicology Screen	Negative for nitrates, organophosphates, heavy metals, HCN
Workshop Participants	84 — Forest Dept + Veterinary Service Dept, CG

15.2 Priority Action Matrix

Priority	Action	Target Area	Timeline
IMMEDIATE	Environmental water/sediment sampling	Rabo & Panikshet dams (RG)	This week
IMMEDIATE	Install earthen ramps at drowning sites	Chhal ponds + Ghargoda dam edge	Dry season
IMMEDIATE	Power-line audit — low-hung lines	All CRITICAL ranges	This month
SEASONAL	Enforcement — illegal crop fencing	All ranges	Oct–Dec annually
SEASONAL	Dry-season calf monitoring	GPS herd watch near water bodies	Jan–May annually
ONGOING	Mandatory PM within 24 hrs + sample dispatch 48 hrs	All ranges	Every event
ANNUAL	EEHV surveillance + systematic documentation	All divisions	Annually
QUARTERLY	Division mortality review meeting	Dharamjaigarh + Raigarh HQ	Every 3 months

Data Sources: Proforma-1 Records, Dharamjaigarh & Raigarh Divisions · PM Reports: ICAR-IVRI Izatnagar (F.2-19/DI/NRC/2025-26/CWL) · GIS Data: Bilaspur Forest Division GIS Cell · Generated: June 2026

Organised by: Chhattisgarh Forest Department · Technical Support: WII, Dehradun · Lab: ICAR-IVRI & NDVSU, Jabalpur